

NETWORK SECURITY ASSESSMENT REPORT

Enterprise Network Segmentation, DMZ Security & Internal Service Protection

Authorized Security Laboratory Assessment
Domain: lab.local • Assessment Date: August 2026
Prepared by: Ishaan Malhotra

Field	Detail
Assessment Type	Authorized Security Laboratory
Primary Focus	Network Security, Firewall Policy, DMZ Segmentation and Security Validation
Core Technologies	pfSense • Ubuntu • Kali Linux • Windows Server 2019 • Windows 10 • Active Directory
Domain	lab.local
Assessment Date	August 2026
Prepared By	Ishaan Malhotra
Overall Result	Effective DMZ segmentation demonstrated

1. Executive Summary

This project implements and assesses an enterprise-style segmented network using pfSense as the perimeter firewall. The environment contains a trusted LAN, an isolated DMZ, and Internet/WAN connectivity. Ubuntu represents a normal DMZ server, Kali Linux represents an attacker and security-testing workstation, and Windows Server 2019 plus Windows 10 represent internal enterprise infrastructure.

The assessment focused on whether a compromised DMZ host could reach internal Active Directory services, perform lateral movement, access the firewall management plane, or bypass segmentation. Testing also verified that legitimate DMZ Internet services remained available.

Key Result

DMZ-to-LAN segmentation was effective. An initial exposure of the pfSense HTTPS management interface from the DMZ was identified, remediated with a specific deny rule, and validated by repeating the exact Nmap test.

2. Assessment Scope

Component	Role	Network / Address
pfSense	Firewall, routing, segmentation, logging	LAN/DMZ gateway
Ubuntu	DMZ server / legitimate-service testing	192.168.20.0/24
Kali Linux	Attacker / security testing	192.168.20.10
Windows Server 2019	Internal Domain Controller + DNS	192.168.10.10
Windows 10	Internal domain workstation	192.168.10.20
Active Directory	Internal identity infrastructure	lab.local

3. Assessment Objectives

- Validate LAN/DMZ segmentation.
- Verify legitimate DMZ Internet connectivity.
- Verify blocked DMZ-to-LAN access to critical Windows/AD services.
- Test firewall management-plane exposure.
- Simulate an SMB lateral-movement attempt.
- Correlate attack/test traffic with pfSense firewall logs.
- Remediate identified exposure and perform post-remediation validation.

4. Network Architecture

The LAN contains the internal Windows infrastructure. The DMZ contains systems that should be treated as less trusted than the LAN. pfSense sits between these zones and enforces explicit traffic policy.

Zone	Subnet	Members
WAN	DHCP (Internet-facing)	pfSense WAN interface
LAN	192.168.10.0/24	Windows Server 2019 (AD DS + DNS), Windows 10 workstation
DMZ	192.168.20.0/24	Ubuntu DMZ server, Kali Linux (security testing)

Security Principle

A compromise of a DMZ system must not automatically become a compromise of internal enterprise infrastructure.

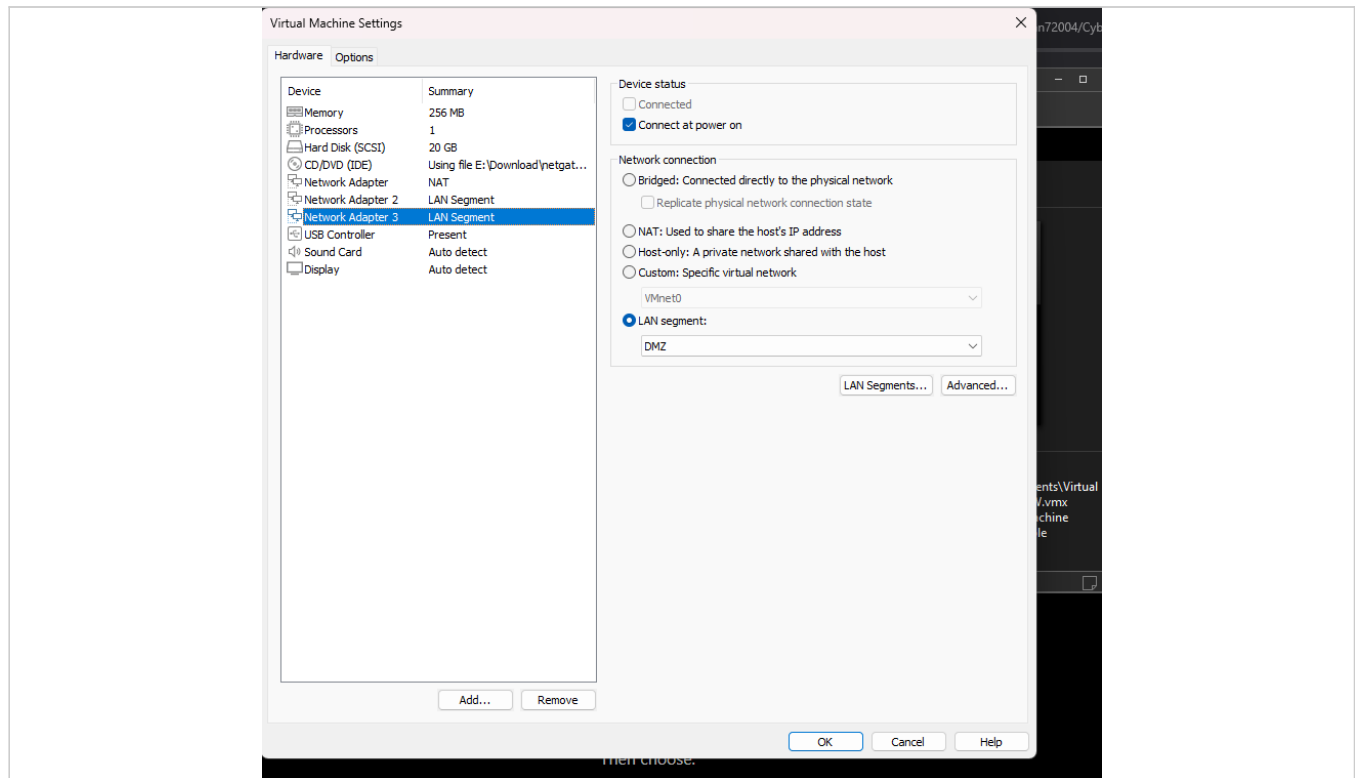


Figure 4.1 — Virtual machine network adapter assigned to the DMZ LAN segment during lab build-out.

5. pfSense Security Policy

Priority	Policy	Purpose
1	ALLOW DMZ → pfSense DNS / UDP 53	Controlled DNS service
2	BLOCK DMZ → LAN / ANY	Primary segmentation control
3	BLOCK DMZ → This Firewall / TCP 443	Protect firewall management plane
4	ALLOW DMZ → Internet / TCP 80	Required HTTP connectivity

Priority	Policy	Purpose
5	ALLOW DMZ → Internet / TCP 443	Required HTTPS connectivity

The specific management-plane deny rule is placed above the general DMZ HTTPS Internet rule. This prevents DMZ access to the firewall itself while preserving normal outbound HTTPS traffic.

5.1 Firewall & Interface Build-Out

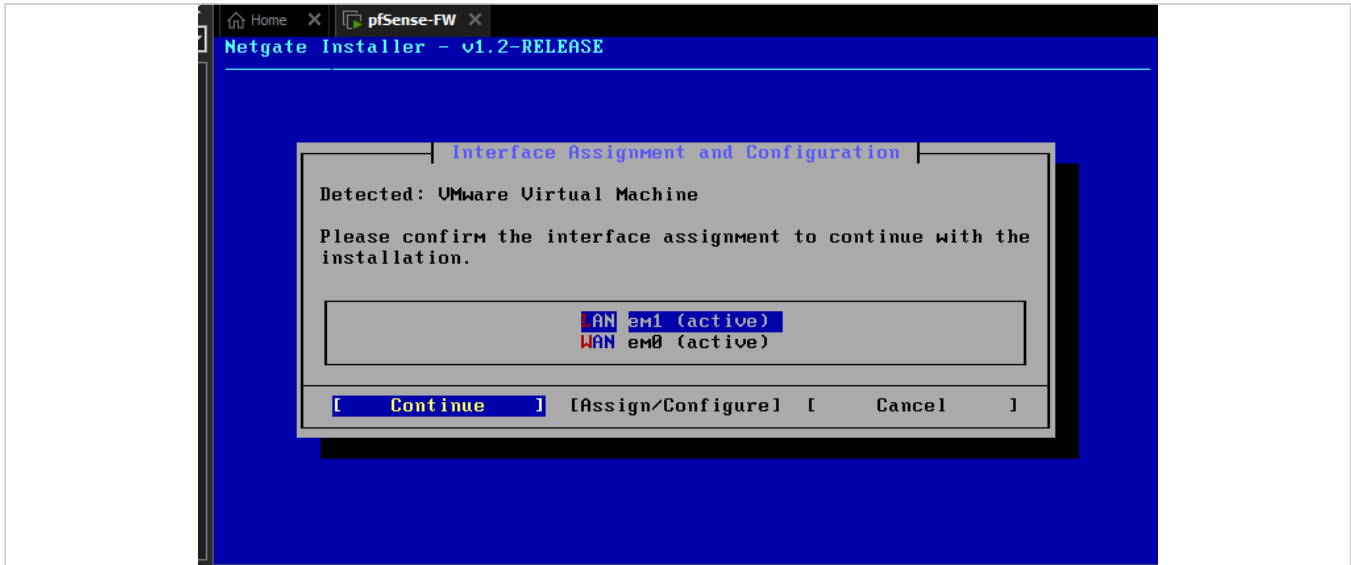


Figure 5.1 — pfSense installer confirming WAN/LAN interface assignment on first boot.

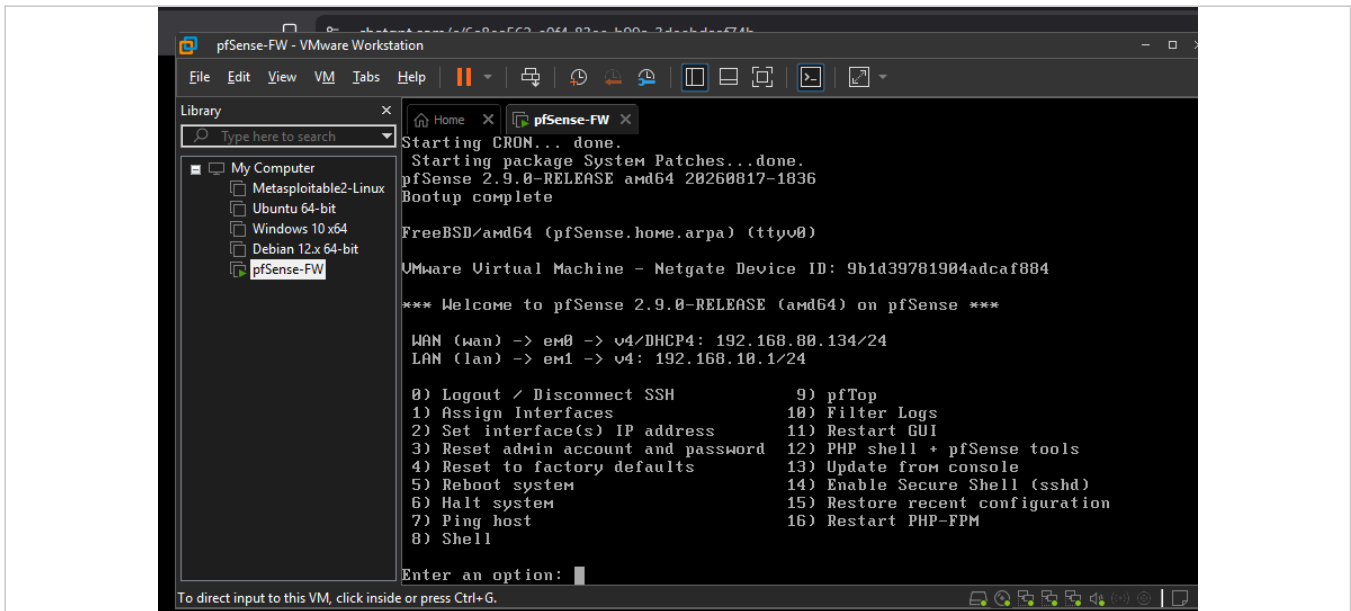


Figure 5.2 — pfSense console after boot, showing WAN (DHCP) and LAN (192.168.10.1/24) assignment.



Figure 5.3 — OPT1 (DMZ) interface configured with IPv4 address 192.168.20.1/24.

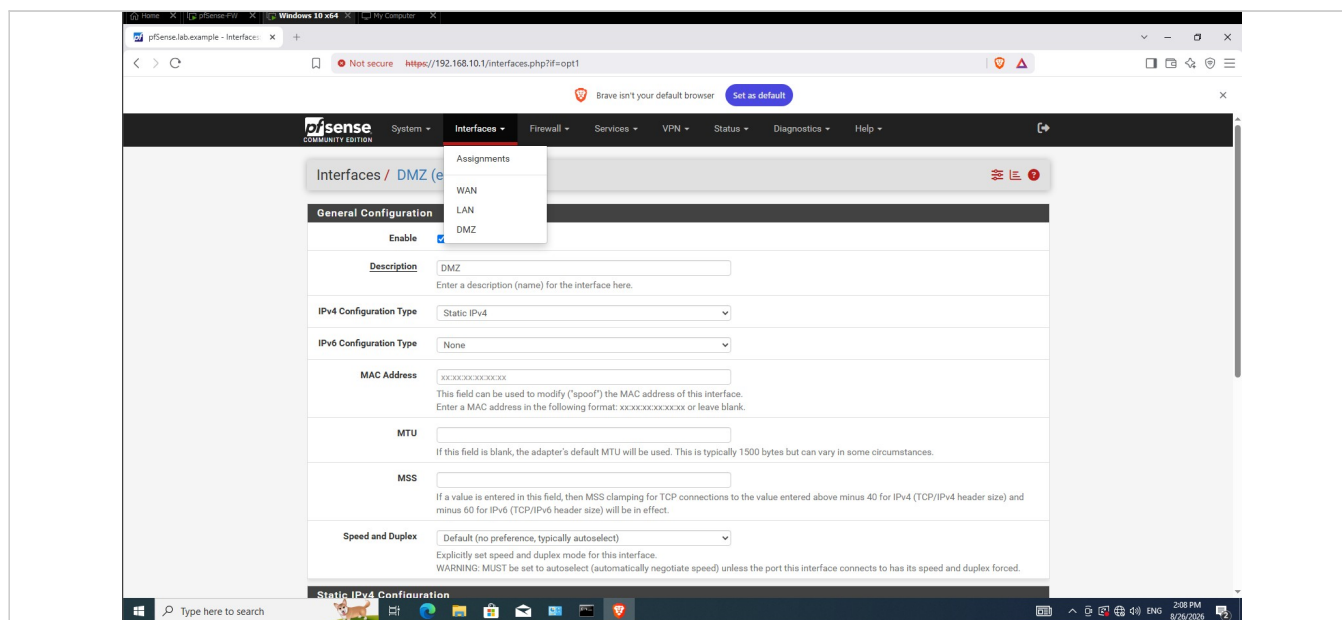


Figure 5.4 — DMZ interface general configuration in the pfSense web GUI.

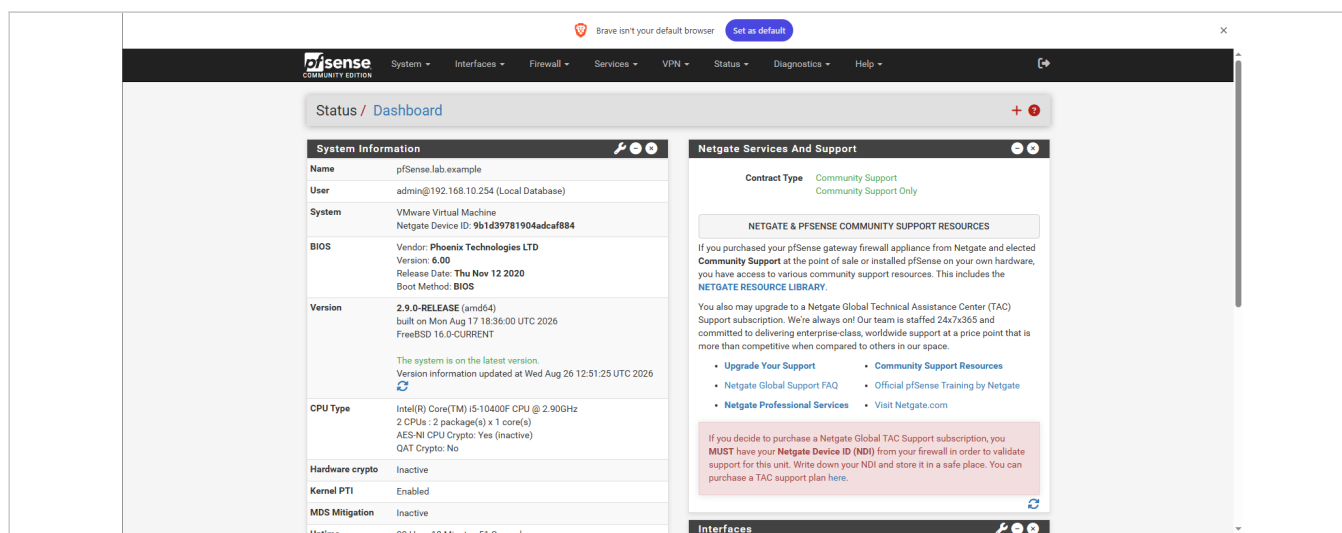


Figure 5.5 — pfSense Status Dashboard confirming system version and uptime after setup.

5.2 Initial DMZ Rule Set

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐		0/0 B	IPv4 UDP	DMZ subnets	*	This Firewall (self)	53 (DNS)	*	none	Allow DMZ DNS	
☐		0/336 B	IPv4 *	DMZ subnets	*	LAN subnets	*	*	none	Block DMZ to LAN	
☐		0/5 KIB	IPv4 *	DMZ subnets	*	*	*	*	none	Allow DMZ to Internet	

Add

Add

Delete

Toggle

Copy

Save

Separator

Figure 5.6 — Initial DMZ firewall rule set: allow DNS, block DMZ-to-LAN, allow DMZ-to-Internet.

5.3 Final DMZ Rule Set (Post-Remediation)

After HTTP/HTTPS were split into individual rules and the firewall management-plane finding (Section 10) was remediated, the DMZ rule base was finalized as shown below.

Firewall / Rules / DMZ

Floating

WAN

LAN

DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐ ✓ ☰	0/22 KiB	IPv4 UDP	DMZ subnets	*	This Firewall (self)	53 (DNS)	*	none		Allow DMZ DNS	✕
☐ ✗ ☰	0/3 KiB	IPv4 *	DMZ subnets	*	LAN subnets	*	*	none		Block DMZ to LAN	✕
☐ ✗ ☰	0/0 B	IPv4 TCP	DMZ subnets	*	This Firewall (self)	443 (HTTPS)	*	none		Block DMZ Firewall HTTPS Management	✕
☐ ✓ ☰	0/33 KiB	IPv4 TCP	DMZ subnets	*	*	80 (HTTP)	*	none		Allow DMZ HTTP	✕
☐ ✓ ☰	0/1.97 MiB	IPv4 TCP	DMZ subnets	*	*	443 (HTTPS)	*	none		Allow DMZ HTTPS	✕

Add

Add

Delete

Toggle

Copy

Save

Separator

Figure 5.7 — Final DMZ rule set, including the dedicated "Block DMZ Firewall HTTPS Management" rule added during remediation.

6. Ubuntu DMZ Security Testing

Ubuntu was used as a DMZ server simulation and was part of the initial firewall validation. The purpose was to verify that legitimate DMZ traffic worked while unnecessary services were restricted.

6.1 – 6.3 Connectivity and Service Restriction

Test	Port	Result
HTTP	TCP/80	Allowed
HTTPS	TCP/443	Allowed
SSH	TCP/22	Blocked

```

ishaan@ubuntu-lab:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f8:83:d5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.20.10/24 brd 192.168.20.255 scope global noprefixroute ens33
        valid_lft forever preferred_lft forever

```

Figure 6.1 — Ubuntu DMZ host configured with static address 192.168.20.10/24 (ens33).

```

ishaan@ubuntu-lab:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=127 time=7.06 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=127 time=8.14 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=127 time=7.07 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=127 time=6.83 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 6.825/7.273/8.144/0.511 ms
ishaan@ubuntu-lab:~$ ping -c 4 google.com
PING google.com (192.178.158.138) 56(84) bytes of data:
64 bytes from lcdels-in-f138.1e100.net (192.178.158.138): icmp_seq=1 ttl=127 time=12.5 ms
64 bytes from lcdels-in-f138.1e100.net (192.178.158.138): icmp_seq=2 ttl=127 time=7.36 ms
64 bytes from lcdels-in-f138.1e100.net (192.178.158.138): icmp_seq=3 ttl=127 time=8.11 ms
64 bytes from lcdels-in-f138.1e100.net (192.178.158.138): icmp_seq=4 ttl=127 time=9.63 ms

```

Figure 6.2 — Ubuntu successfully reaching 1.1.1.1 and google.com, confirming outbound Internet connectivity.

```

ishaan@ubuntu-lab:~$ ping -c 4 192.168.10.254
PING 192.168.10.254 (192.168.10.254) 56(84) bytes of data:

--- 192.168.10.254 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3066ms

ishaan@ubuntu-lab:~$

```

Figure 6.3 — Ubuntu ping to LAN gateway (192.168.10.254) — 100% packet loss, confirming DMZ-to-LAN isolation.

```

ishaan@ubuntu-lab:~$ timeout 5 bash -c '</dev/tcp/example.com/80' && echo "HTTP 80 OPEN" || echo "HTTP 80 BLOCKED"
HTTP 80 OPEN
ishaan@ubuntu-lab:~$ timeout 5 bash -c '</dev/tcp/example.com/443' && echo "HTTPS 443 OPEN" || echo "HTTPS 443 BLOCKED"
HTTPS 443 OPEN
ishaan@ubuntu-lab:~$ timeout 5 bash -c '</dev/tcp/example.com/22' && echo "SSH 22 OPEN" || echo "SSH 22 BLOCKED"
SSH 22 BLOCKED
ishaan@ubuntu-lab:~$

```

Figure 6.4 — Direct TCP connect tests from Ubuntu: HTTP 80 OPEN, HTTPS 443 OPEN, SSH 22 BLOCKED — matching the configured DMZ policy exactly.

✗	Aug 26 15:57:17	DMZ	Default deny rule IPv4 (1000000103)	192.168.20.10:37502	172.66.147.243:22	TCP-S
✓	Aug 26 15:57:17	DMZ	Allow DMZ DNS (1787758539)	192.168.20.10:45179	192.168.20.1:53	UDP
✓	Aug 26 15:57:18	DMZ	Allow DMZ HTTP (1787750578)	192.168.20.10:35278	185.125.190.101:80	TCP-S
✗	Aug 26 15:57:18	DMZ	Default deny rule IPv4 (1000000103)	192.168.20.10:37502	172.66.147.243:22	TCP-S
✗	Aug 26 15:57:19	DMZ	Default deny rule IPv4 (1000000103)	192.168.20.10:37502	172.66.147.243:22	TCP-S

Figure 6.5 — Correlating pfSense firewall log: SSH attempt hits the default deny rule while HTTP/DNS traffic is explicitly allowed.

States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/0 B	IPv4 UDP	DMZ subnets	*	This Firewall (self)	53 (DNS)	*	none	Allow DMZ DNS	[Add] [Edit] [Delete] [Toggle] [Copy] [Save] [Separator]
✗	0/672 B	IPv4 *	DMZ subnets	*	LAN subnets	*	*	none	Block DMZ to LAN	[Add] [Edit] [Delete] [Toggle] [Copy] [Save] [Separator]
✓	0/0 B	IPv4 TCP	DMZ subnets	*	*	80 (HTTP)	*	none	Allow DMZ HTTP	[Add] [Edit] [Delete] [Toggle] [Copy] [Save] [Separator]
✓	0/0 B	IPv4 TCP	DMZ subnets	*	*	443 (HTTPS)	*	none	Allow DMZ HTTPS	[Add] [Edit] [Delete] [Toggle] [Copy] [Save] [Separator]

Figure 6.6 — Refined DMZ rule set separating HTTP (80) and HTTPS (443) into explicit allow rules.

6.4 Ubuntu Security Model

Ubuntu represents a normal DMZ server that could be compromised in a real environment. The security objective is to ensure that such a compromise does not provide direct access to internal LAN infrastructure.

The screenshot displays the pfSense firewall log interface. The log table contains 20 entries. The first 10 entries show allowed DMZ traffic to the Internet (rule 1787750578) with green checkmarks. The next 6 entries show blocked DMZ traffic to the LAN (rule 1787749945) with red X marks. The final 4 entries show allowed DMZ traffic to the Internet (rule 1787750578) with green checkmarks. The log is viewed on a Windows desktop with the taskbar at the bottom showing the time as 4:42 PM on 8/26/2020.

Time	Action	Rule	Source IP	Destination IP	Protocol
Aug 26 15:27:17	✓	Allow DMZ to Internet (1787750578)	192.168.20.10:33942	91.189.91.98:80	TCP-S
Aug 26 15:28:57	✓	Allow DMZ to Internet (1787750578)	192.168.20.10:37151	185.125.190.56:123	UDP
Aug 26 15:31:10	✗	Block DMZ to LAN (1787749945)	192.168.20.10	192.168.10.254	ICMP
Aug 26 15:31:11	✗	Block DMZ to LAN (1787749945)	192.168.20.10	192.168.10.254	ICMP
Aug 26 15:31:12	✗	Block DMZ to LAN (1787749945)	192.168.20.10	192.168.10.254	ICMP
Aug 26 15:31:13	✗	Block DMZ to LAN (1787749945)	192.168.20.10	192.168.10.254	ICMP
Aug 26 15:32:17	✓	Allow DMZ to Internet (1787750578)	192.168.20.10:44178	192.168.20.1:53	TCP-S
Aug 26 15:32:18	✓	Allow DMZ to Internet (1787750578)	192.168.20.10:34150	185.125.190.101:80	TCP-S
Aug 26 15:33:14	✓	Allow DMZ to Internet (1787750578)	192.168.20.10:33746	185.125.190.56:123	UDP
Aug 26 15:37:17	✓	Allow DMZ to Internet (1787750578)	192.168.20.10:44974	192.168.20.1:53	TCP-S
Aug 26 15:37:18	✓	Allow DMZ to Internet (1787750578)	192.168.20.10:48136	91.189.91.96:80	TCP-S
Aug 26 15:38:41	✗	Block DMZ to LAN (1787749945)	192.168.20.10	192.168.10.254	ICMP
Aug 26 15:38:42	✗	Block DMZ to LAN (1787749945)	192.168.20.10	192.168.10.254	ICMP
Aug 26 15:38:43	✗	Block DMZ to LAN (1787749945)	192.168.20.10	192.168.10.254	ICMP
Aug 26 15:38:44	✗	Block DMZ to LAN (1787749945)	192.168.20.10	192.168.10.254	ICMP
Aug 26 15:39:19	✓	Allow DMZ to Internet (1787750578)	192.168.20.10	1.1.1.1	ICMP
Aug 26 15:39:51	✓	Allow DMZ to Internet (1787750578)	192.168.20.10:52210	192.168.20.1:53	TCP-S
Aug 26 15:39:51	✓	Allow DMZ to Internet (1787750578)	192.168.20.10	192.178.158.101	ICMP

Figure 6.7 — pfSense firewall log showing repeated "Block DMZ to LAN" entries alongside allowed outbound DMZ Internet traffic.

7. Internal Windows Infrastructure

7.1 Windows Server 2019

Windows Server 2019 was placed on the LAN with a static address of 192.168.10.10. The server was promoted to an Active Directory Domain Controller and provided DNS for the internal domain lab.local.

Setting	Value
IP address	192.168.10.10
Subnet mask	255.255.255.0
Default gateway	192.168.10.1
Domain	lab.local
Roles	AD DS + DNS

```

Administrator: Command Prompt

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::4c97:d3e0:3751:15a7%3
    IPv4 Address. . . . . : 192.168.10.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.1

C:\Users\Administrator>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Administrator>ping 1.1.1.1

Pinging 1.1.1.1 with 32 bytes of data:
Reply from 1.1.1.1: bytes=32 time=6ms TTL=127
Reply from 1.1.1.1: bytes=32 time=5ms TTL=127
Reply from 1.1.1.1: bytes=32 time=6ms TTL=127
Reply from 1.1.1.1: bytes=32 time=4ms TTL=127
  
```

Figure 7.1 — Windows Server 2019 ipconfig output (192.168.10.10/24) with successful gateway and Internet reachability tests.

7.2 Active Directory

Active Directory was deployed to provide a realistic internal enterprise target. The AD infrastructure included OUs, users, and security groups. AD is supporting infrastructure for the network-security assessment rather than the primary IAM objective.

Example security groups included IT-Admin, Network-Admins, Soc-Analysts, and Helpdesks.

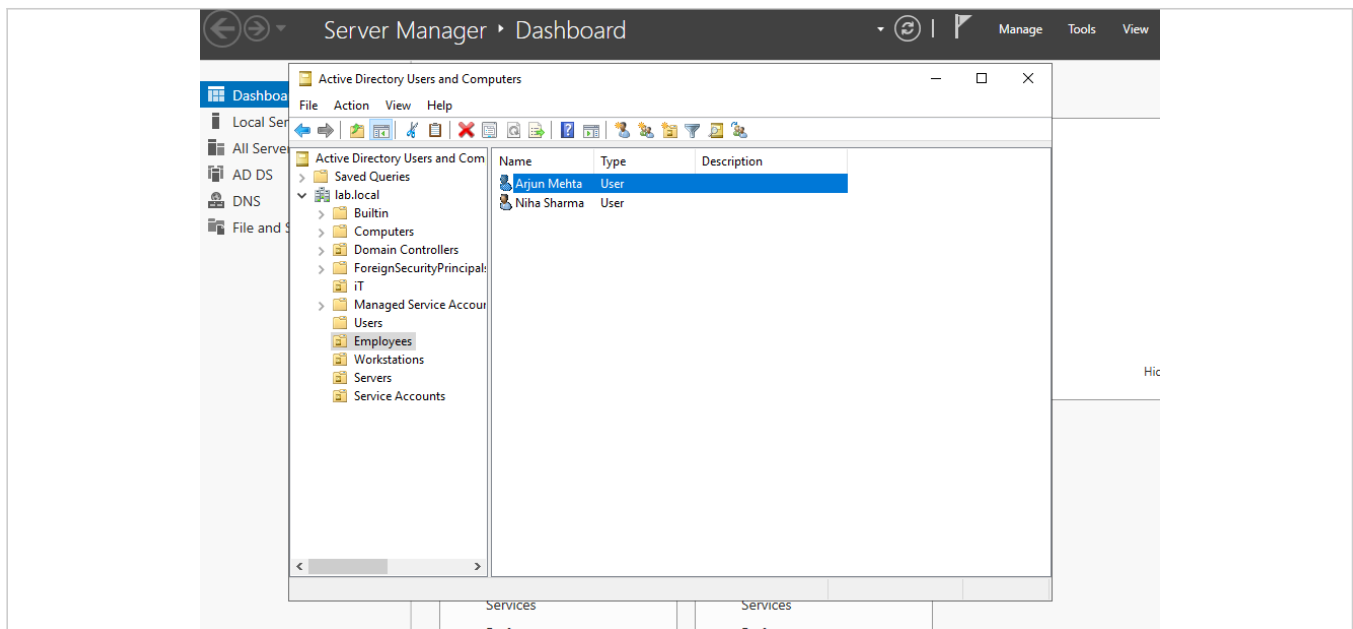


Figure 7.2 — Active Directory Users and Computers, showing the Employees OU with domain user accounts (Arjun Mehta, Niha Sharma).

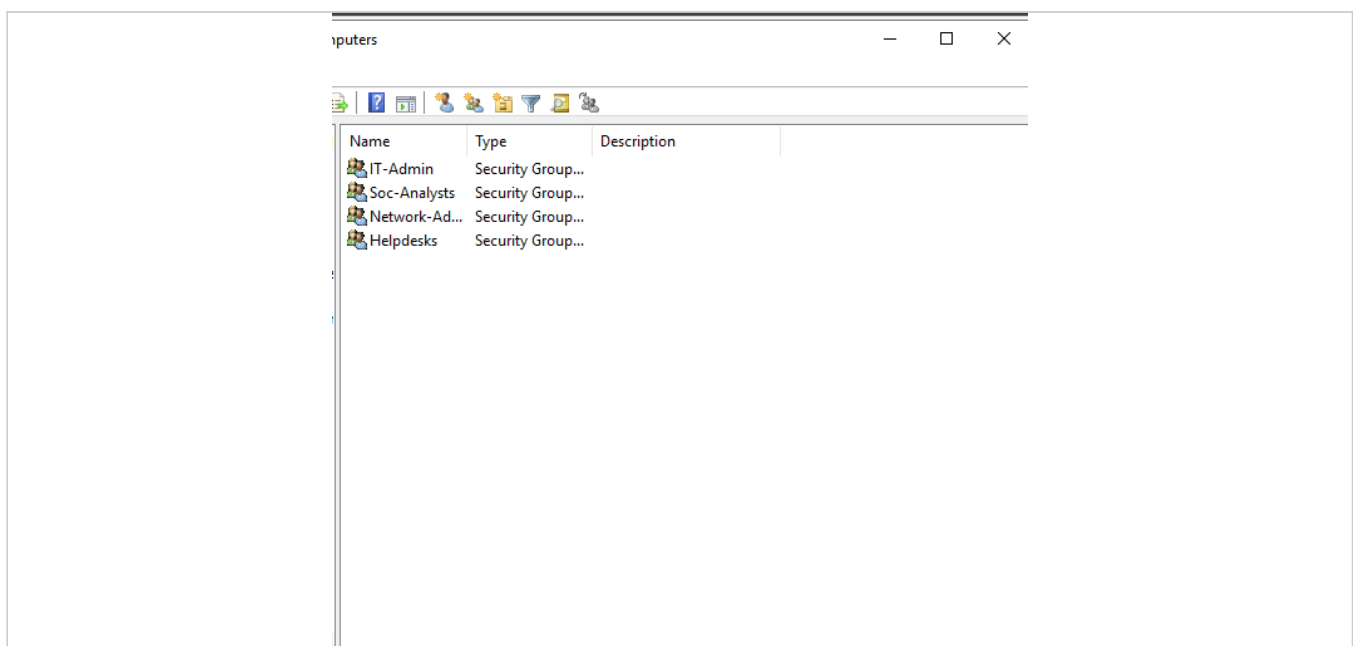


Figure 7.3 — Security groups configured in Active Directory: IT-Admin, Soc-Analysts, Network-Admins, Helpdesks.

7.3 Windows 10 Workstation

The Windows 10 workstation was placed on the LAN at 192.168.10.20, configured to use 192.168.10.10 for DNS, and successfully joined to lab.local.

Domain authentication was validated using an AD account, confirming that the internal Windows environment was functioning as a realistic enterprise network asset.

```

Administrator: Command Prompt

Reply from 192.168.20.10: bytes=32 time=1ms TTL=63
Reply from 192.168.20.10: bytes=32 time=1ms TTL=63

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Windows\system32>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Windows\system32>nslookup lab.local
Server:      Unknown
Address:     192.168.10.10

Name:       lab.local
Address:    192.168.10.10

C:\Windows\system32>

```

Figure 7.4 — Windows 10 connectivity to the DMZ and Domain Controller, plus successful nslookup resolution of lab.local.

```

Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Arjun-Mehta>whoami
lab\arjun-mehta

C:\Users\Arjun-Mehta>whoami /groups

GROUP INFORMATION
-----

```

Group Name	Type	SID	Attributes
Everyone	Well-known group	S-1-1-0	Mandatory group
p, Enabled by default, Enabled group	Alias	S-1-5-32-545	Mandatory group
BUILTIN\Users	Well-known group	S-1-5-4	Mandatory group
NT AUTHORITY\INTERACTIVE	Well-known group	S-1-2-1	Mandatory group
p, Enabled by default, Enabled group	Well-known group	S-1-5-11	Mandatory group
CONSOLE LOGON	Well-known group	S-1-5-15	Mandatory group
NT AUTHORITY\Authenticated Users	Well-known group	S-1-2-0	Mandatory group
p, Enabled by default, Enabled group	Well-known group	S-1-5-21-3254936382-736894326-3095553212-1104	Mandatory group
NT AUTHORITY\This Organization	Well-known group		
p, Enabled by default, Enabled group	Well-known group		
LOCAL	Well-known group		
p, Enabled by default, Enabled group	Well-known group		
LAB\Soc-Analysts	Group		

Figure 7.5 — whoami /groups confirming authenticated domain session as LAB\arjun-mehta with Soc-Analysts group membership.

8. Kali Linux Security Testing

Kali Linux was used as the attacker/security-testing workstation in the DMZ. This was deliberately different from Ubuntu: Ubuntu represented a normal DMZ server, while Kali represented a hostile or security-testing endpoint.

8.1 DMZ → LAN Reconnaissance

The Windows Domain Controller at 192.168.10.10 was tested for common Windows/AD services.

Port	Service	Security Relevance
53	DNS	AD name resolution
88	Kerberos	AD authentication / attack path
135	RPC	Windows remote services
139	NetBIOS	Legacy Windows networking
389	LDAP	Directory services
445	SMB	Windows file sharing / lateral movement
3389	RDP	Remote administration

Nmap results showed the internal services as filtered, demonstrating that the firewall prevented direct DMZ-to-LAN service access.

```
(kali㉿kali)-[~]
└─$ nmap -Pn -sT -p 53,88,135,139,389,445,3389 192.168.10.10
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 12:48 -0700
Nmap scan report for 192.168.10.10
Host is up.

PORT      STATE      SERVICE
53/tcp    filtered  domain
88/tcp    filtered  kerberos-sec
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
389/tcp   filtered  ldap
445/tcp   filtered  microsoft-ds
3389/tcp  filtered  ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 3.52 seconds
```

Figure 8.1 — Nmap scan of the Domain Controller from Kali: all seven target ports (53, 88, 135, 139, 389, 445, 3389) return "filtered."

✗	Aug 26 19:45:11	DMZ	Default deny rule IPv4 (1000000103)	i 192.168.20.10:59075	i 192.46.210.39:123	UDP
✗	Aug 26 19:45:21	DMZ	Default deny rule IPv4 (1000000103)	i 192.168.20.10:34832	i 192.46.211.253:123	UDP
✓	Aug 26 19:48:26	DMZ	Allow DMZ DNS (1787758539)	i 192.168.20.10:39088	i 192.168.20.1:53	UDP
✗	Aug 26 19:48:27	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:38952	i 192.168.10.10:53	TCP:S
✗	Aug 26 19:48:27	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:44458	i 192.168.10.10:3389	TCP:S
✗	Aug 26 19:48:27	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:56684	i 192.168.10.10:135	TCP:S
✗	Aug 26 19:48:27	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:50188	i 192.168.10.10:88	TCP:S
✗	Aug 26 19:48:27	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:59516	i 192.168.10.10:389	TCP:S
✗	Aug 26 19:48:27	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:47392	i 192.168.10.10:139	TCP:S
✗	Aug 26 19:48:27	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:33798	i 192.168.10.10:445	TCP:S
✗	Aug 26 19:48:29	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:59518	i 192.168.10.10:389	TCP:S
✗	Aug 26 19:48:29	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:33810	i 192.168.10.10:445	TCP:S
✗	Aug 26 19:48:29	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:56686	i 192.168.10.10:135	TCP:S
✗	Aug 26 19:48:29	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:44462	i 192.168.10.10:3389	TCP:S
✗	Aug 26 19:48:29	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:50204	i 192.168.10.10:88	TCP:S
✗	Aug 26 19:48:29	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:38962	i 192.168.10.10:53	TCP:S
✗	Aug 26 19:48:29	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:47406	i 192.168.10.10:139	TCP:S

Figure 8.2 — pfSense firewall log correlating the exact Nmap scan above, showing each probe hit by the "Block DMZ to LAN" rule.

8.2 DMZ → AD DNS

Command: `dig @192.168.10.10 lab.local`

The query timed out and returned no reachable DNS server. This demonstrated that the DMZ could not directly query the Domain Controller's DNS service.

```
(kali㉿kali)-[~]
└─$ dig @192.168.10.10 lab.local
;; communications error to 192.168.10.10#53: timed out
;; communications error to 192.168.10.10#53: timed out
;; communications error to 192.168.10.10#53: timed out

; <<>> DiG 9.20.23-1-Debian <<>> @192.168.10.10 lab.local
; (1 server found)
;; global options: +cmd
;; no servers could be reached
```

Figure 8.3 — dig query to the Domain Controller's DNS service times out from the DMZ.

8.3 – 8.6 Individual Service Timeouts

```
(kali㉿kali)-[~]
└─$ nc -vz -w 3 192.168.10.10 445
192.168.10.10: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.10.10] 445 (microsoft-ds) : Connection timed out
```

Figure 8.4 — SMB / TCP 445: nc connection timed out.

```
(kali㉿kali)-[~]
└─$ nc -vz -w 3 192.168.10.10 3389
192.168.10.10: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.10.10] 3389 (ms-wbt-server) : Connection timed out
```

Figure 8.5 — RDP / TCP 3389: nc connection timed out.

```
(kali㉿kali)-[~]  
$ nc -vz -w 3 192.168.10.10 389  
192.168.10.10: inverse host lookup failed: Unknown host  
(UNKNOWN) [192.168.10.10] 389 (ldap) : Connection timed out
```

Figure 8.6 — LDAP / TCP 389: nc connection timed out.

```
(kali㉿kali)-[~]  
$ nc -vz -w 3 192.168.10.10 88  
192.168.10.10: inverse host lookup failed: Unknown host  
(UNKNOWN) [192.168.10.10] 88 (kerberos) : Connection timed out
```

Figure 8.7 — Kerberos / TCP 88: nc connection timed out.

9. Attack Simulation — SMB Lateral Movement

A controlled SMB lateral-movement attempt was performed from the DMZ using a lab domain account. The goal was to determine whether an attacker-controlled DMZ host could reach the Domain Controller's SMB service.

Command: `smbclient -L //192.168.10.10 -U 'LAB\Arjun-Mehta'`

Observed result: NT_STATUS_IO_TIMEOUT. pfSense simultaneously recorded traffic from 192.168.20.10 to 192.168.10.10:445 under the Block DMZ to LAN policy.

Outcome

The attack failed before SMB authentication could occur. This is strong evidence that the network boundary stopped the lateral-movement path.

```
(kali㉿kali)-[~]
$ smbclient -L //192.168.10.10 -U 'LAB\Arjun-Mehta'
o_connect: Connection to 192.168.10.10 failed (Error NT_STATUS_IO_TIMEOUT)
```

Figure 9.1 — `smbclient` SMB enumeration attempt against the Domain Controller returns NT_STATUS_IO_TIMEOUT.

✗	Aug 26 20:09:57	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:36316	i 192.168.10.10:3389	TCP:S
✓	Aug 26 20:11:39	DMZ	Allow DMZ DNS (1787758539)	i 192.168.20.10:53073	i 192.168.20.1:53	UDP
✗	Aug 26 20:11:39	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:59480	i 192.168.10.10:389	TCP:S
✗	Aug 26 20:11:40	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:59480	i 192.168.10.10:389	TCP:S
✗	Aug 26 20:11:41	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:59480	i 192.168.10.10:389	TCP:S
✗	Aug 26 20:13:52	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:34936	i 192.168.10.10:445	TCP:S
✗	Aug 26 20:13:52	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:32998	i 192.168.10.10:139	TCP:S
✗	Aug 26 20:13:53	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:32998	i 192.168.10.10:139	TCP:S
✗	Aug 26 20:13:53	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:34936	i 192.168.10.10:445	TCP:S
✗	Aug 26 20:13:54	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:34936	i 192.168.10.10:445	TCP:S
✗	Aug 26 20:13:54	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:32998	i 192.168.10.10:139	TCP:S
✗	Aug 26 20:13:55	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:32998	i 192.168.10.10:139	TCP:S
✗	Aug 26 20:13:55	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:34936	i 192.168.10.10:445	TCP:S
✗	Aug 26 20:13:56	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:34936	i 192.168.10.10:445	TCP:S
✗	Aug 26 20:13:56	DMZ	Block DMZ to LAN (1787749945)	i 192.168.20.10:32998	i 192.168.10.10:139	TCP:S

Figure 9.2 — pfSense firewall log correlating the SMB lateral-movement attempt, blocking DMZ traffic to 192.168.10.10 on ports 445/139/3389/389.

9.1 Kerberoasting Attack-Path Validation

A direct Kerberos connectivity check was performed against TCP/88. The connection timed out, demonstrating that the DMZ could not directly reach the Domain Controller's Kerberos service.

Scope Limitation

Kerberoasting itself was not executed. No service ticket was extracted or cracked. The test demonstrates network-level prevention of the Kerberoasting path.

10. Security Finding — Firewall Management Exposure

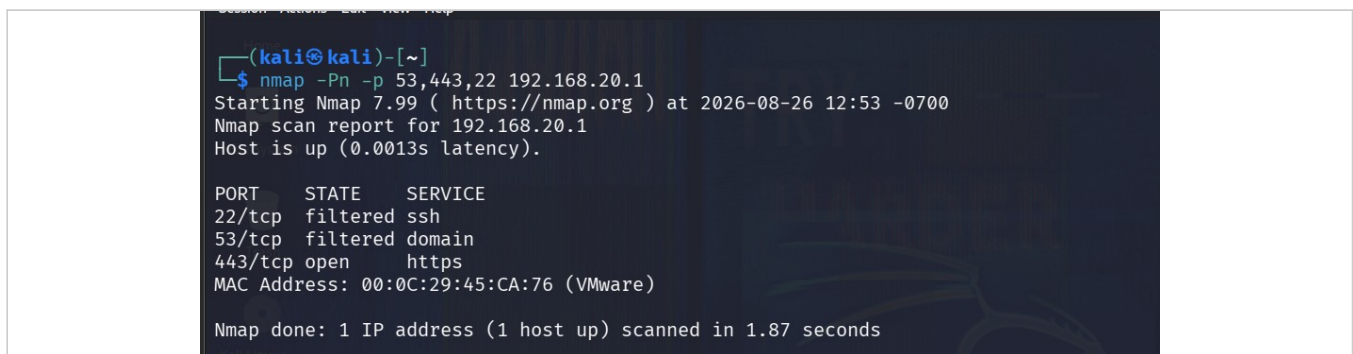
Finding N-01

The pfSense HTTPS management interface was initially reachable from the DMZ.

10.1 Discovery

Command: `nmap -Pn -p 53,443,22 192.168.20.1`

Initial result: 443/tcp open https. SSH was filtered and TCP/53 was filtered in the corresponding test. The exposed HTTPS management service represented unnecessary management-plane reachability.



```
(kali㉿kali)-[~]
└─$ nmap -Pn -p 53,443,22 192.168.20.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 12:53 -0700
Nmap scan report for 192.168.20.1
Host is up (0.0013s latency).

PORT      STATE      SERVICE
22/tcp    filtered  ssh
53/tcp    filtered  domain
443/tcp   open      https
MAC Address: 00:0C:29:45:CA:76 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.87 seconds
```

Figure 10.1 — Initial Nmap scan of the pfSense firewall from the DMZ: port 443/tcp shows OPEN.

10.2 Risk

If a DMZ host were compromised, unnecessary access to the firewall management plane could increase the attack surface and provide an additional route toward infrastructure control.

10.3 Remediation

Setting	Remediation
Action	BLOCK
Protocol	TCP / IPv4
Source	DMZ subnets
Destination	This Firewall
Destination port	HTTPS / 443
Description	Block DMZ Firewall HTTPS Management

The deny rule was placed above the general DMZ HTTPS Internet rule. This preserves normal outbound HTTPS access while blocking HTTPS access to the firewall itself.

10.4 Validation

State	Result
Before	443/tcp OPEN

State	Result
After	443/tcp FILTERED

The exact same Nmap test was repeated after the rule was applied. The state change from open to filtered confirms successful remediation.

```
(kali㉿kali)-[~]  
$ nmap -Pn -p 53,443,22 192.168.20.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 13:02 -0700  
Nmap scan report for 192.168.20.1  
Host is up (0.00053s latency).  
  
PORT      STATE      SERVICE  
22/tcp    filtered  ssh  
53/tcp    filtered  domain  
443/tcp    filtered  https  
MAC Address: 00:0C:29:45:CA:76 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.85 seconds
```

Figure 10.2 — Post-remediation Nmap scan: port 443/tcp now returns FILTERED, confirming the fix.

11. Assessment Methodology

The project followed an evidence-driven security testing lifecycle. Tests were performed from the relevant zone, results were correlated with pfSense logs, and identified weaknesses were remediated and re-tested.

Discover	Test	Analyze	Remediate	Validate
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Evidence-driven security assessment lifecycle

12. Complete Security Test Matrix

ID	Source	Target	Test	Result	Status
T01	Ubuntu DMZ	Internet	HTTP/80	Allowed	PASS
T02	Ubuntu DMZ	Internet	HTTPS/443	Allowed	PASS
T03	Ubuntu DMZ	Service	SSH/22	Blocked	PASS
T04	Kali DMZ	pfSense	SSH/22	Filtered	PASS
T05	Kali DMZ	pfSense	DNS/53	Filtered by policy	PASS
T06	Kali DMZ	pfSense	HTTPS/443	Initially open	FINDING
T07	Kali DMZ	pfSense	HTTPS/443 after fix	Filtered	PASS
T08	Kali DMZ	Windows DC	DNS/53	Timeout	PASS
T09	Kali DMZ	Windows DC	Kerberos/88	Timeout	PASS
T10	Kali DMZ	Windows DC	RPC/135	Filtered	PASS
T11	Kali DMZ	Windows DC	NetBIOS/139	Filtered	PASS
T12	Kali DMZ	Windows DC	LDAP/389	Timeout	PASS
T13	Kali DMZ	Windows DC	SMB/445	Timeout	PASS
T14	Kali DMZ	Windows DC	RDP/3389	Timeout	PASS
T15	Kali DMZ	Windows DC	SMB lateral movement	NT_STATUS_IO_TIMEOUT	PASS
T16	Kali DMZ	Windows DC	Kerberoasting path / 88	Blocked	PASS

13. Evidence Register

The following evidence is included in this portfolio report. The strongest evidence pairs an action/test with the corresponding firewall enforcement or before/after result.

Evidence	Purpose	Figure
Ubuntu HTTP/HTTPS test	Demonstrates legitimate DMZ Internet service access	6.2, 6.4
Ubuntu SSH blocked test	Demonstrates restricted DMZ service policy	6.4
Windows Server ipconfig + gateway ping	Demonstrates LAN connectivity	7.1
AD/DNS validation	Demonstrates internal enterprise infrastructure	7.2, 7.3, 7.4
Windows 10 domain join/authentication	Demonstrates realistic internal workstation	7.4, 7.5
Kali Nmap → Windows DC	Demonstrates internal service isolation	8.1
Kali → AD DNS timeout	Demonstrates DNS isolation	8.3
Kali → SMB timeout + pfSense log	Demonstrates SMB segmentation	8.4, 9.2
Kali → RDP timeout	Demonstrates RDP isolation	8.5
Kali → LDAP timeout	Demonstrates LDAP isolation	8.6
Kali → Kerberos timeout	Demonstrates Kerberos isolation	8.7
Initial pfSense Nmap 443 open	Evidence of discovered finding	10.1
pfSense deny-rule (final rule set)	Evidence of remediation	5.7
Post-remediation Nmap 443 filtered	Evidence of successful validation	10.2
SMB lateral-movement attempt + firewall log	Strong attack simulation evidence	9.1, 9.2

14. Risk Summary

Finding	Severity	Status	Remediation
DMZ access to pfSense HTTPS management	Medium	Remediated	Block DMZ → This Firewall TCP/443
Direct DMZ access to AD DNS	Controlled	Passed	Existing DMZ → LAN block
Direct DMZ access to SMB	Controlled	Passed	Existing DMZ → LAN block
Direct DMZ access to RDP	Controlled	Passed	Existing DMZ → LAN block
Direct DMZ access to LDAP	Controlled	Passed	Existing DMZ → LAN block
Direct DMZ access to Kerberos	Controlled	Passed	Existing DMZ → LAN block

15. Final Assessment

The lab successfully demonstrates enterprise network segmentation and security validation using pfSense. The Ubuntu phase established that legitimate DMZ services could function while unnecessary traffic such as SSH was restricted. The Windows Server and Windows 10 systems provided realistic internal enterprise infrastructure, including Active Directory and DNS.

Kali Linux was then used to simulate an attacker operating from the DMZ. The attacker could not directly reach the Domain Controller's DNS, Kerberos, LDAP, SMB, or RDP services. An SMB lateral-movement attempt produced NT_STATUS_IO_TIMEOUT and was confirmed in the pfSense firewall logs.

The only significant exposure identified was the initial ability of the DMZ to reach the pfSense HTTPS management interface. This was remediated with a dedicated firewall rule and independently validated by repeating the Nmap scan.

Overall Assessment

EFFECTIVE DMZ SEGMENTATION DEMONSTRATED.

16. Portfolio Summary

This project demonstrates practical skills in firewall configuration, network segmentation, DMZ architecture, service-level filtering, Nmap reconnaissance, Netcat validation, DNS testing, SMB security testing, lateral-movement simulation, firewall log analysis, remediation, and post-remediation validation.

The strongest portfolio narrative is:

Narrative Arc

Architecture → Configure → Discover → Attack/Test → Detect → Remediate → Re-test → Document

All tests described in this report were performed within an authorized isolated laboratory environment.

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